TOP-Q — Day 11 worked notes

Quantum extension: Hong–Ou–Mandel interference

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Scope kept small on purpose (risk R9): two photons, one 2×2 device. No multi-photon mesh. All results are derived or **simulated**; no photons were measured.

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1 The HOM effect

Send one photon into each input of a 50:50 beam splitter. If the two photons are identical in every respect (colour, arrival time, shape, polarisation), they *always leave together* — the probability of one photon at each output (a "coincidence") is zero. This is Hong, Ou & Mandel, *Phys. Rev. Lett.* **59**, 2044 (1987), your reference [7] [Literature]. Making the photons different (e.g. delaying one) brings coincidences back: plotting coincidences against delay gives the **HOM dip**.

In simple words: two identical photons meeting at a 50:50 splitter "bunch": they leave by the same door. The more different they are, the more often they leave by different doors.

2 Derivation

2.1 Identical photons, any splitting ratio [Derived; Verified-by-test]

Use the coupler of Day 8 for the photon operators: $a^\dagger \to \sqrt{\eta}\, c^\dagger + i\sqrt{1-\eta}\, d^\dagger$, $b^\dagger \to i\sqrt{1-\eta}\, c^\dagger + \sqrt{\eta}\, d^\dagger$. The input $a^\dagger b^\dagger |0\rangle$ becomes

$$\left(\sqrt{\eta}\, c^\dagger + i\sqrt{1-\eta}\, d^\dagger\right)\left(i\sqrt{1-\eta}\, c^\dagger + \sqrt{\eta}\, d^\dagger\right)|0\rangle.$$

The $c^\dagger d^\dagger$ (coincidence) term has amplitude $\eta + i^2(1-\eta) = 2\eta - 1$: the two ways for the photons to end up in different outputs (both pass straight, or both cross) *cancel*. So $P_c = (2\eta-1)^2 = (T-R)^2$, zero at 50:50.

Independent check: a direct calculation in the two-mode Fock space (matrices for a, b; $U = e^{iH}$) agrees to 10^{-16}.

2.2 Partly different photons [Derived; Verified-by-test]

If the photons' internal states are ρ_1 and ρ_2 (pure: $|\psi_1\rangle, |\psi_2\rangle$), only the overlapping part interferes. For any 2×2 device T:

$$P_c = |T_{11}T_{22}|^2 + |T_{12}T_{21}|^2 + 2\,\mathrm{Re}\big(T_{11}T_{22}\overline{T_{12}T_{21}}\big)\,\mathrm{Tr}(\rho_1\rho_2).$$

For a beam splitter, $P_c = R^2 + T^2 - 2RT\,\mathrm{Tr}(\rho_1\rho_2)$. The **visibility** (dip depth relative to fully distinguishable photons) is

$$\boxed{V = \frac{2RT}{R^2+T^2}\,\mathrm{Tr}(\rho_1\rho_2)} \qquad (= |\langle\psi_1|\psi_2\rangle|^2 \text{ at 50:50 for pure states}).$$

Checked against an explicit symmetrised two-photon calculation with random 4-dimensional internal states and a random lossless device (agreement 10^{-12}).

2.3 Three ways to make photons different

- **Delay** τ (Gaussian spectra, rms width σ_ω): $|J|^2 = e^{-\sigma_\omega^2\tau^2}$ — this gives the dip shape.
- **Colour mismatch** $\Delta\omega$: extra factor $e^{-\Delta\omega^2/4\sigma_\omega^2}$.
- **Spatial-mode mismatch**: a factor $|\int u_1^* u_2\, dA|^2$ (overlap of the two output mode shapes; e.g. photons leaving a waveguide in slightly different modes).

3 Distinguishability is not the same as decoherence

Distinguishability: each photon is in a definite (pure) state, but the two states differ ($\rho_1 \neq \rho_2$). Example: a fixed colour mismatch. **Decoherence / impurity**: each photon's state is mixed — it is a random one of several possibilities, shot to shot. Example: random timing jitter. Even if both photons come from identical sources ($\rho_1 = \rho_2 = \rho$), $V = \mathrm{Tr}(\rho^2) =$ the **purity**, which is below 1 [Derived].

Can one HOM measurement tell them apart? The visibility alone cannot: both only enter through $\mathrm{Tr}(\rho_1\rho_2)$. In our Gaussian models, the *dip shape* can (Fig. 1b): a colour mismatch chosen to give $V = 0.5$ leaves the dip width unchanged (FWHM 1.64 in units of $1/\sigma_\omega$), while timing jitter chosen to give the same $V = 0.5$ **doubles** the width (FWHM 3.32) [Simulated]. In general, separating the two needs extra measurements (e.g. dip shape, or interfering photons from independent sources to measure purity). We state this as a limitation, not a solved problem.

4 Connection to TOP-Q

4.1 A single beam splitter does not care about phase

A phase on one input before a 50:50 splitter does not change V at all (checked over 0–2π: change $< 3 \times 10^{-16}$) [Derived; Simulated]. Two-photon interference at one splitter depends on the **splitting ratio** and the **overlap**, not on any interferometric phase. So TOP-Q's phase protection matters for HOM only when the "splitter" is itself an interferometer.

4.2 Coupler gap errors

With $\eta = \frac{1}{2} + \delta$: $V \approx 1 - 8\delta^2$. Using Day 8, $\delta = -\frac{\pi}{4}\epsilon_\kappa$: $\boxed{V \approx 1 - \frac{\pi^2}{2}\sigma_\kappa^2}$. Monte Carlo over random couplers agrees for $\sigma_\kappa \leq 0.05$ (e.g. 0.9957 vs. 0.9956 at $\sigma_\kappa = 0.03$) and is slightly better than the formula for larger errors. Couplers are not topological; no array design changes this.

4.3 An MZI used as the 50:50 splitter, with array arms [Simulated]

On a chip, a tunable 50:50 splitter is an MZI set to $\theta = \pi/2$. Its arms now matter: an arm-to-arm phase error changes the effective splitting ratio, and an arm amplitude imbalance does too. Using the Day 9–10 arm elements (4000 MZIs each):

Mean HOM visibility	T	D	W (plain guide)
gap errors $\sigma_\kappa = 0.01$	0.9998	0.975	1 (exact)
gap errors $\sigma_\kappa = 0.03$	0.998	0.835	1 (exact)
width errors $\sigma_\beta = 0.01\kappa_2$	0.947	0.942	0.917
width errors $\sigma_\beta = 0.03\kappa_2$	0.701	0.692	0.618

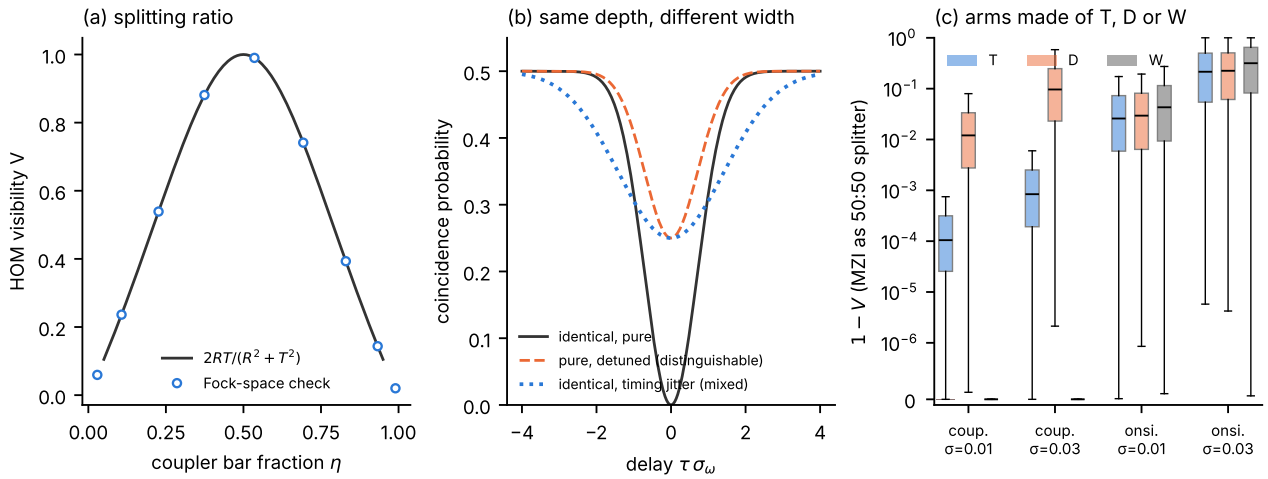


Figure 1: **Simulation**, `scripts/day11_hom.py` → `data/day11_hom.npz`. (a) Visibility vs. coupler splitting ratio; circles: independent Fock-space calculation. (b) HOM dips: identical pure photons; pure but colour-detuned photons; identical photons with timing jitter (mixed state), both chosen to give $V = 0.5$. (c) Spread of $1-V$ for an MZI-based 50:50 splitter whose arms are T, D or W elements (boxes: 25–75%; whiskers 5–95%).

The same pattern as the classical circuit: T keeps the quantum interference against gap errors (much better than D, but no better than a plain waveguide) and gives no protection against width errors. Note how quickly HOM visibility collapses with width errors in arms 10 coupling lengths long — the 5% worst MZIs drop to $V \approx 0.16$ at $\sigma_\beta = 0.03\kappa_2$.

5 What the HOM model can and cannot represent

Can: splitting-ratio errors, arm phase and amplitude errors inside an MZI splitter, pure-state distinguishability (delay, colour, spatial mode), mixed states via $\mathrm{Tr}(\rho_1\rho_2)$. **Cannot (needs a fuller quantum-optical model):** multi-photon interference in a full mesh (permanents of $N \times N$ matrices), photon loss with non-post-selected statistics, imperfect sources (multi-pair emission), detector efficiency and dark counts, time-dependent noise during propagation.

6 Hardware: how HOM is measured

Not done by us. The images in our chat (published work) show on-chip and fibre HOM setups: a photon-pair source (often a nonlinear crystal or a silicon ring), a variable delay line, the beam splitter (bulk, fibre or on-chip coupler), and two single-photon detectors connected to a coincidence counter. The measured coincidence rate vs. delay is the dip of Fig. 1b. Cite the source papers shown with each image before reusing a photo.

7 Code and tests

New: `topq/hom.py`; `scripts/day11_hom.py`; `tests/test_hom.py` (10 tests, including the Fock-space and first-quantised independent checks). Suite: **138 passed**.

8 Self-test

1. Derive $P_c = (2\eta - 1)^2$ for identical photons.
2. Why does a phase on one input not change the HOM dip?
3. What is the difference between a distinguishable photon pair and a mixed (impure) photon pair? Why can V alone not tell them apart?
4. Show $V \approx 1 - \frac{\pi^2}{2}\sigma_\kappa^2$ for coupler gap errors.
5. Why do the arm elements matter when the splitter is an MZI?